

Qingbiao Li

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Work Experience

Assistant Professor

Macao, P.R China

Faculty of Science and Technology, University of Macau

Aug 2024 - Present

Research Direction: 1) Robot Learning for multi-robot and multi agent coordination, LLMs-based autonomous agent;
2) Machine Learning for intra-operative imaging for surgical robotics, medical imaging and diagnosis.

Postdoctoral Researcher for RAILS

Oxford, UK

Oxford Robotics Institute (Supervisor: Dr Lars Kunze)

Feb 2023 - May 2024

Focus on investigating causal inference for corner case generation for autonomous driving.

Research Intern in Project Silica

Cambridge, UK

Microsoft Research Cambridge (Supervisor: Dr Ioan Stefanovici & Dr Katja Hofmann)

July 2021 - Oct 2021

Explore explainable RL-based approaches to scheduling in the Silica glass library, towards a scheduler for production deployment.

Education

University of Cambridge

Cambridge, UK

PhD in Computer Science (Supervisor: **Prof Amanda Prorok**)

Oct 2018 - Dec 2022

- **Research Interest:** Robot Learning, Multi-robot Path Planning, On-device Learning, Graph Neural Networks (GNNs), Imitation Learning, Reinforcement Learning and Computer Vision (Medical Imaging).
- **Programming Languages:** Python, PyTorch, Tensorflow, PyTorch Geometric, Jax, Jraph, Deep Graph Library, C++.
- **Framework and Simulator:** Unity ML-Agent, Pybullet, Mujoco, Gazebo, AWS Robomaker, RLlib, ROS2 and ROS.

Imperial College London

London, UK

MRes Medical Robotics and Image Guided Intervention (Distinction)

Oct 2017 - Sep 2018

- **Master Thesis** (Supervisor: **Prof Daniel Elson**): Developed a tissue oxygenation monitoring technique using multispectral imaging and conditional generative adversarial networks (cGANs)

The University of Edinburgh

Edinburgh, UK

M. Eng (Hons) Mechanical Engineering (Thesis Supervisor: **Dr Filipe Teixeira-Dias**)

Sep 2013 - June 2016

- **MEng Thesis:** Missile impact on snow inspired by British Antarctic Survey's project.

South China University of Technology

Guangzhou, China

B. Eng. Mechanical Engineering and Automation

Sep 2011 - July 2013

Research Experience

ROBOTICS

Graph Neural Networks for Decentralized Multi-robot Path Planning

Cambridge, UK

Prorok Lab, University of Cambridge (Supervisor: Dr Amanda Prorok)

Oct 2018 - Dec 2022

- The **first** to use graph neural networks (GNNs) for explicit communication between a cooperative multirobot team for motion planning.
- Efficient, collision-free navigation for thousands of agents, using our Message-Aware Graph Attention Networks (MA-GATs): **video**
- Sim2Real for reinforcement learning to navigate robot team through a narrow passage in continuous motion: **video**

Research Assistant in Bipedal Walking of Humanoid Robot

Edinburgh, UK

SLMC, The University of Edinburgh (Supervisors: Dr Zhibin Li & Prof Sethu Vijayakumar)

Sep 2016 - June 2017

- Robust control for bipedal locomotion using online Tikhonov regularisation: **video**.

COMPUTER VISION (MEDICAL IMAGING)

Real-time Surgical Environment Enhancement for Robot-Assisted MIS

London, UK

Hamlyn Centre, Imperial College London (Supervisor: Dr Benny Lo)

Mar 2020 - Sep 2020

- Multi-scale super-resolution Generative Adversarial Network (GAN) for Robot-Assisted Minimally Invasive Surgery.
- Co-supervised Master student, provided academic guidance, revised paper and iterated it as ICRA 2021 paper: **video**

Vision-based Navigation in Flexible Endoscopy

London, UK

Hamlyn Centre, Imperial College London (Supervisor: Dr George Mylonas)

Sep 2017 - Dec 2017

- Customised multiple visual-inertial SLAM methods for endoscope use within the human body: **video**.

Publications

JOURNAL ARTICLES

- Bo Lu, Tiancheng Zhou, **Qingbiao Li**, Bin Li, Jiewen Lai, Yu Wang, Yunhui Liu, Lining Sun, Peng Qi. MIGUEL-Net: A Monocular Image-Guided Depth Localization Network for Robotic Navigation in Endoscopic Submucosal Dissection *IEEE Transactions on Medical Imaging (JCR Q1, IF 8.9)* (Under Review). Elsevier, 2024
- Kangyu Ji, Weizhe Lin, Yuqi Sun, Lin-Song Cui, Javad Shamsi, Yu-Hsien Chiang, Jiawei Chen, Elizabeth M Tennyson, Linjie Dai, **Qingbiao Li**, Kyle Frohna, Miguel Anaya, Neil C. Greenham, Samuel D. Stranks. Self-supervised deep learning for tracking degradation of perovskite light-emitting diodes with multispectral imaging *Nature Machine Intelligence (JCR Q1, IF 23.8)* pp. 1–11. Nature Publishing Group UK London, 2023
- Linhao Yang, Bidan Huang, **Qingbiao Li**, Ya-Yen Tsai, Wang Wei Lee, Chaoyang Song, Jia Pan. TacGNN: Learning Tactile-based In-hand Manipulation with a Blind Robot using Hierarchical Graph Neural Network *IEEE Robotics and Automation Letters (JCR Q2, IF 5.2)*. IEEE, 2023
- Jiajun Cao*, **Qingbiao Li***, Liping Xu, Rui Yang, Yuejin Dai. “Non-Parametric Surrogate Model Method Based on Machine Learning With Application on Low-Pressure Steam Turbine Exhaust System,” *Journal of the Global Power and Propulsion Society (JCR Q3, IF 1.209)*. 2021. **PDF**
- Qingbiao Li**, Weizhe Lin, Zhe Liu, Amanda Prorok. “Message-Aware Graph Attention Networks for Large-Scale Multi-Robot Path Planning,” *IEEE Robotics and Automation Letters (JCR Q2, IF 5.2)*. 2020. **PDF**
- Fernando Gama, **Qingbiao Li**, Ekaterina Tolstaya, Amanda Prorok, Alejandro Ribeiro. “Synthesizing Decentralized Controllers with Graph Neural Networks and Imitation Learning,” *IEEE Transactions on Signal Processing (JCR Q1, IF 5.4)*. 2022. **PDF**
- Binyu Wang, Zhe Liu, **Qingbiao Li**, Amanda Prorok. “Mobile Robot Path Planning in Dynamic Environments through Globally Guided Reinforcement Learning,” *IEEE Robotics and Automation Letters (JCR Q2, IF 5.2)* pp. 6932–6939. 2020. **PDF**
- Weizhe Lin, Indigo Orton, **Qingbiao Li**, Gabriela Pavarini, Marwa Mahmoud. “Looking At The Body: Automatic Analysis of Body Gestures and Self-Adaptors in Psychological Distress,” *IEEE Transactions on Affective Computing (JCR Q1, IF 11.2)*. Springer, 2020. **PDF**
- Qingbiao Li**, Jianyu Lin, Neil T Clancy, Daniel S Elson. “Estimation of Tissue Oxygen Saturation from RGB Images and Sparse Hyperspectral Signals based on Conditional Generative Adversarial Network,” *International Journal of Computer Assisted Radiology and Surgery (JCR Q2, IF 3)*. pp. 987–995. Springer, 2019. **PDF**

CONFERENCE PROCEEDINGS

- Huanrong Liu, Chunlin Tian, Xuyang Wei, Jiaheng Dai, Qin Liu, Tianqi Wei, **Qingbiao Li**, Li Li. RAP: Runtime-Adaptive Pruning for LLM Inference *Conference on Neural Information Processing Systems (NeurIPS) (CCF-A)* (Under Review). 2025
- Chuang Wu, Youzhan Wang, Xiaozheng Li, **Qingbiao Li**, Krishna Manaswi Digumarti, Chongjing Cao**. Design and Characterization of a Thermal-electrostatic Dual-modal Soft Pouch Motor *IEEE/RSJ International Conference on Intelligent Robots and System (IROS) (CCF-C, Qualis-A1)*. 2025
- Youzhan Wang, Jinjun Li, Chuang Wu, Xiaozheng Li, **Qingbiao Li**, Krishna Manaswi Digumarti, Chongjing Cao. T-Touch: a Soft Thermal-haptic Multimodal Fingertip Wearable Device for Immersive Virtual Reality *IEEE/RSJ International Conference on Intelligent Robots and System (IROS) (CCF-C, Qualis-A1)*. 2025
- Yong Ma, Chuang Wu, **Qingbiao Li**, Chongjing Cao, Xiaozheng L. A Bio-inspired Stiffness-programmable Robotic Flexible Joint Based on Electro-adhesive Clutches *IEEE/RSJ International Conference on Intelligent Robots and System (IROS) (CCF-C, Qualis-A1)*. 2025
- Chenning Yu*, **Qingbiao Li***, Gao Sicun, Amanda Prorok. Accelerating Multi-Agent Planning using Graph Transformers with Near-Optimal Guarantees *IEEE International Conference on Robotics and Automation (CCF-B, Qualis-A1)*. 2023
- Benjamin Hudson, **Qingbiao Li**, Matthew Malencia, Amanda Prorok. “Graph Neural Network Guided Local Search for the Traveling Salesperson Problem,” *International Conference on Learning Representations (CCF-A, Qualis-A1)*, 2022, **PDF**
- Jan Blumenkamp *, **Qingbiao Li***, Binyu Wang, Zhe Liu, Amanda Prorok. See What the Robot Can’t See: Learning Cooperative Perception for Visual Navigation *IEEE/RSJ International Conference on Intelligent Robots and System (CCF-C, Qualis-A1)*. 2023. **PDF**
- Jan Blumenkamp *, **Qingbiao Li***, Binyu Wang, Zhe Liu, Amanda Prorok. Learning to Navigate using Visual Sensor Networks *IEEE International Conference on Robotics and Automation (CCF-B, Qualis-A1)*. 2023. **PDF**, **best paper at CoPerception Workshop**
- Jan Blumenkamp, Steven Morad, Jennifer Gielis, **Qingbiao Li**, Amanda Prorok. “A Framework for Real-World Multi-Robot Systems Running Decentralized GNN-Based Policies,” *IEEE International Conference on Robotics and Automation (CCF-B, Qualis-A1)*, 2021
- Lifeng Zhou, Vishnu D Sharma, **Qingbiao Li**, Amanda Prorok, Alejandro Ribeiro, Pratap Tokekar, Vijay Kumar. Graph neural networks for decentralized multi-robot submodular action selection *IEEE International Conference on Safety, Security, and Rescue Robotics*. 2021. **PDF**
- Amanda Prorok, Jan Blumenkamp, **Qingbiao Li**, Ryan Kortvelesy, Zhe Liu, Ethan Stump. “The Holy Grail of Multi-Robot Planning: Learning to Generate Online-Scalable Solutions from Offline-Optimal Experts,” *International Conference on Autonomous Agents and MultiAgent Systems (CCF-A)*. 2022. **PDF**
- Jan Blumenkamp, **Qingbiao Li**, Amanda Prorok. “Evaluating the Sim-to-Real Gap of Graph Neural Network Policies for Multi-Robot Coordination,” *IEEE International Conference on Robotics and Automation (CCF-B, Qualis-A1)*, *Real World Swarms Workshop*, 2021, **PDF**
- Qingbiao Li**, Fernando Gama, Alejandro Ribeiro, Amanda Prorok. “Graph Neural Networks for Decentralized Multi-robot Path Planning,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (CCF-C, ERA-A, Qualis-A1)*, 2020, **PDF**, **Publication of the Year**
- Qingbiao Li**, Fernando Gama, Alejandro Ribeiro, Amanda Prorok. “Graph Neural Networks for Decentralized Path Planning,” *International Conference on Autonomous Agents and MultiAgent Systems (CCF-A, ERA-A, Qualis-A1)*, 2020, **PDF**
- Ruoxi Wang, Dandan Zhang, **Qingbiao Li**, Xiao-Yun Zhou, Benny Lo. “Real-time Surgical Environment Enhancement for Robot-Assisted Minimally Invasive Surgery Based on Super-Resolution,” *IEEE International Conference on Robotics and Automation (CCF-B, Qualis-A1)*, 2021, **PDF**
- Qingbiao Li**, Xiao-Yun Zhou, Jianyu Lin, Jian-Qing Zheng, Neil T Clancy, Daniel S Elson. “Estimation of Tissue Oxygen Saturation from RGB Images based on Pixel-level Image Translation,” *The Hamlyn Symposium on Medical Robotics*, 2018, **PDF**
- Qingbiao Li**, Iordanis Chatzinikolaïdis, Yiming Yang, Sethu Vijayakumar, Zhibin Li. “Robust Foot Placement Control for Dynamic Walking using Online Parameter Estimation,” *IEEE-RAS 17th International Conference on Humanoid Robotics (Humanoids) (CCF-C, Qualis-A1)*, 2017, **PDF**

Funding

Deep Learning based Decentralized Heterogeneous Embodied Intelligence Robot Collaboration

1,900k MOP,
2025-2027

Science and Technology Development Fund, Macau SAR

Efficient Embodied Intelligence for Robotics Locomotion and Control	250k RMB, 2025-2026
Avant Robotics Co., Ltd. (Industrial Funded Project)	
Automated Data Collection and Annotation via Ensemble Learning for Medical Imaging and Embodied Intelligence Dataset	150k MOP, 2024-2026
Starting Research Grant, University of Macau	
Grasping and Manipulation Demonstration with Bi-Manipulators Based on Robot Learning Technology	150k RMB, 2024-2025
Shenzhen Moying Technology Co., Ltd. (Industrial Funded Project)	

Honors & Awards

Chinese Government Award for Outstanding Self-financed Students Abroad	2023
China Scholarship Council	
Chui Wen Mei Memorial Bursary	2022
Hughes Hall, University of Cambridge	
Wiseman Prize	2020
Department of Computer Science and Technology, University of Cambridge	
Subsystem Excellence Award at Hyperloop Pod Competition	2016
Space Exploration Technologies Corporation	
International Student Scholarship	2013-2016
The University of Edinburgh	

Teaching Experience

Course Coordinator for Undergraduate Embodied Robotics Intelligence Degree	Macau, P.R. China
Dept of Electromechanical Engineering, University of Macau	
6th Jan 2025	
- Oversaw the design, implementation, and continuous improvement of the undergraduate robotics curriculum including robotics and machine learning towards Embodied Robotics Intelligence Degree. - Coordinated course content, schedules, and assessments to ensure alignment with program objectives and industry standards. - Collaborated with faculty to integrate interdisciplinary concepts and practical applications into the program. - Ensured quality assurance and maintained academic standards in line with institutional policies.	
Lecturer for Undergraduate Robotics Course	Macau, P.R. China
Dept of Electromechanical Engineering, University of Macau	
6th Jan 2025	
- Designed and delivered a comprehensive course covering key topics in robotics, including robot terminology, spatial transformations, manipulator kinematics, Jacobians, trajectory generation, and control strategies (linear and non-linear). - Facilitated student learning through practical examples, interactive lectures, and hands-on projects to strengthen conceptual understanding and technical skills.	
Guest Lecturer for Introduction to Deep Learning	London, UK
I-X initiative, Imperial College London	
5th Dec 2024	
Designed two-hour lecture about the introduction to Graph Neural Networks and its application, designed practical CoLab coding demonstration to enhance student's understanding about Graph Neural Networks.	
Guest Lecturer for Introduction to Deep Learning	Macau, P.R. China
Faculty of Science and Technology, University of Macau	
11th Nov 2024	
Designed three-hour lecture about the introduction to Graph Neural Networks and its application, designed practical tutorial to enhance student's understanding about Graph Neural Networks.	

Guest Lecturer for Motion Capture for Experimental Robotics Research

Cambridge, UK

Department of Computer Science and Technology, University of Cambridge

Jan 2019 - Dec 2022

- Designed two-hour lecture that includes a one-hour lecture on the fundamentals and advancements of motion capture technology in robotics, followed by a one-hour live demonstration focusing on trajectory following real robots and its the practical application to enhance the precision and performance of trajectory following.

Teaching Assistant / Lab Demonstrator in Multi-Robot Systems (MRS)

Cambridge, UK

Department of Computer Science and Technology, University of Cambridge

Jan 2019 - Dec 2022

- Provided Q and A session with students about practical assignments, and supervised students' mini-project.
- Simulation environment ROS and AWS RoboMaker, physical experiment using TurtleBot.

Teaching Assistant

Edinburgh, UK

School of Engineering, The University of Edinburgh

Jan 2016 - Nov 2016

- Mechanical Engineering 1**
 - Basic of Statics and Dynamics, Solid Mechanics and Thermodynamics.
- Fluid Mechanics 2**
 - Fluid Statics, Bernoulli's Equation, Hydraulic Structures.

Supervision Experience

MPhil Thesis Supervision

Macau, P.R.China

Faculty of Science and Technology, University of Macau

Sep 2024 - Present

- Tiantian Zhang, Tinghuang, Chen Tong, Shiyuan Yang, Zhijia Tao, Chan Cheok Hin

MPhil Thesis Supervision

Cambridge, UK

Department of Computer Science and Technology, University of Cambridge

Sep 2020 - Dec 2023

- Peter Woo, 2022-2023
- Benjamin Hudson & George Papagiannis, 2020-2021

Invited Talks

Machine Learning for Embodied Artificial Intelligence: from Surgical Robotics to Multi-robot Coordination

Sep. 2024

Department of Computer Science and Engineering, The Chinese University of Hong Kong (CUHK)

Graph Neural Network for Multi-robot Coordination

Jan. 2024

Institute for Interdisciplinary Information Sciences, Tsinghua University

Graph Neural Network Guided Local Search for the Traveling Sales-person Problem

Dec. 2022

Lightning Talk at Learning on Graphs, University of Cambridge

Message-Aware Graph Attention Networks for Large-Scale Multi-Robot Path Planning

April. 2021

University of Pennsylvania, Philadelphia, United States

From Graph Neural Networks to Decentralized Multi-Robot Path Planning

Dec. 2020

Zhejiang University, Hangzhou, China

Graph Neural Networks for Decentralized Path Planning

Dec. 2019

Robotics X Tencent, Shenzhen, China

Skills and Language Proficiency

Courses and Software MOOC Certificate, AutoCAD, PTC Creo, Microsoft Office, DaVinci Resolve, MATLAB, $\text{\LaTeX}$ Chinese Mandarin (Native), Cantonese (Intermediate)

English Proficient

German Basic (Passed A2)

Community Activities

Program Committee

2022-Present

AAMAS 2023 Blue-Sky Track, AAAI 2022

Journal/Conference Reviewer

2017-Present

IJRR, T-RO, Autonomous Robots (AURO), RA-L, IROS, ICRA, RA-L, AAMAS, RSS

Community Activities

Contributing to Chinese Documentation of Deep Graph Library (GDL)

2020

Department of Computer Science and Technology, University of Cambridge